

# Screenshots

# RLE

### Visual File Compression Playground

Interactive OS Course Project - File I/O + Memory Visualization

Upload File

No file selected

Size: 0 B

Type: -

Download Compressed File

RLE

Huffman

LZ77

Compare All

AAAAABBBBBCCCCDDDDDDDDDD

#### How RLE Works

Run-Length Encoding replaces consecutive repeated characters with a count and the character. "AAAAABBB" becomes "4A3B".

21

Original Chars

4

Encoded Tokens

0.190

Compression Ratio

A

A

A

A

B

B

C

C

C

C

C

D

D

D

D

D

D

D

D

D

Encoded Output:

4

A

2

B

5

C

10

D

#### Memory Tracker

|                |             |        |
|----------------|-------------|--------|
| Input Buffer   | <div></div> | 42 B   |
| Working Memory | <div></div> | 10.5 B |
| Output Buffer  | <div></div> | 16 B   |

# Huffman Coding

Upload File

No file selected Size: 0 B Type: -

Download Compressed File

RLE

Huffman

LZ77

Compare All

abccdeehhhhh

How Huffman Works  
Builds a binary tree based on character frequency. Common characters get shorter codes, rare ones get longer codes.

12  
Characters

28  
Compressed Bits

0.292  
Compression Ratio

```
graph TD; Root(( )) --- b((b)); Root --- h((h)); b --- d((d)); b --- e((e)); e --- a((a)); e --- c((c));
```

d  
000  
freq: 2

e  
001  
freq: 2

b  
0100  
freq: 1

c  
0101  
freq: 1

a  
011  
freq: 1

h  
1  
freq: 5

011 0100 0101 000 000 001 001 1 1 1 1 1

Memory Tracker

Input Buffer

Working Memory (Tree)

Output Buffer

24 B

96 B

3.5 B

LZ77

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No file selected

Size: 0 B

Type: -

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RLE

Huffman

LZ77

Compare All

Lamborghini Lamborghini

How LZ77 Works

Uses a sliding window to find repeated patterns. Instead of repeating data, it writes (offset, length, next\_char) triples pointing back to earlier occurrences.

23

Original Chars

13

Tokens

0.652

Compression Ratio

Step: 1 / 13

Buffer (History)

L

Lookahead

amborghini Lamborghini

Lit: 'L'

Lit: 'a'

Lit: 'm'

Lit: 'b'

Lit: 'o'

Lit: 'n'

Lit: 'g'

Lit: 'h'

Lit: 'i'

Lit: 'n'

Lit: 'i'

Lit: '.'

(12, 11, '')

Memory Tracker

Input Buffer

46 B

Working Memory (Window)

92 B

Output Buffer

234 B

# Comparison

